



Laboratory Work 6

H-infinite Optimal Controller design

Lab Goal: To design an H_∞ optimal controller design for the disturbed linear plant

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1. Introduction

In this laboratory work we will design an H_∞ optimal controller design for the disturbed linear plant in the form,

$$\dot{x} = Ax + Bu + B_f f$$

We have to synthesize an optimal controller using the following Riccati equation,

$$A^T P + PA + Q - PBB^T P + \gamma^{-2} P B_f B_f^T P = 0$$

$$K = -B^T * P$$

Let's first calculate the minimum value of γ for which the semi positive defined matrix P exists as the Riccati equation solution,

By using the Riccati equation solver on matlab and experimentally changing the value of γ we have found out that at $\gamma = 15.685$ is the minimum value at which the eigen values of the system are negative so that the system is stable at this point and the matrix P exists as the solution for the given Riccati equation

Moreover we have also calculated the controller $u = Kx$ from the solution of P by using the given equation,

We will run the plant on the initial conditions $[1 \ 0]^T$ and the external disturbance,

$$f = 10\sin 6t + 5\cos 2t + 4\cos 3t + 3\cos 8t$$

We have the data as following,

$$A = \begin{bmatrix} 3 & -1 \\ -2 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 8 \\ 8 \end{bmatrix}$$

$$Q = \begin{bmatrix} 9 & 0 \\ 0 & 7 \end{bmatrix}, \quad Bf = \begin{bmatrix} 0 \\ 7 \end{bmatrix}$$

2. Modeling

Let's model the system now on matlab simulink and plot the required outputs for the minimum value of gamma,

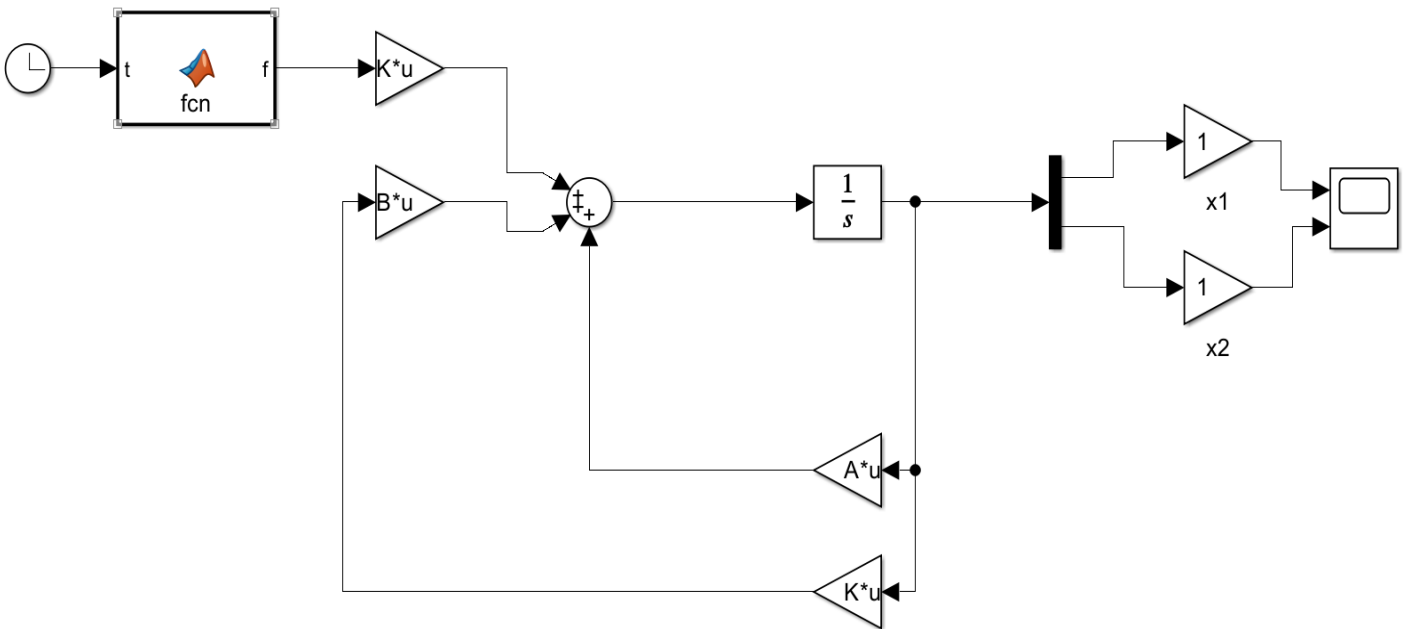


Figure 1. System Model

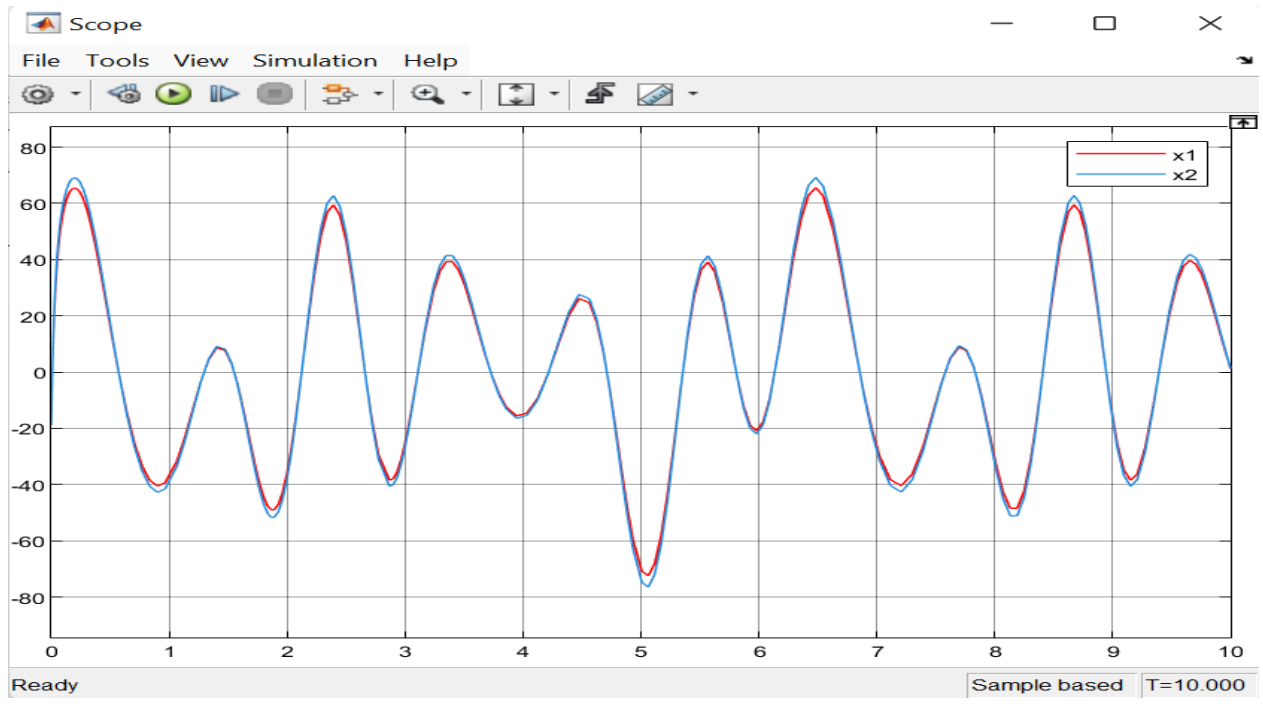


Figure 2. States x_1 and x_2

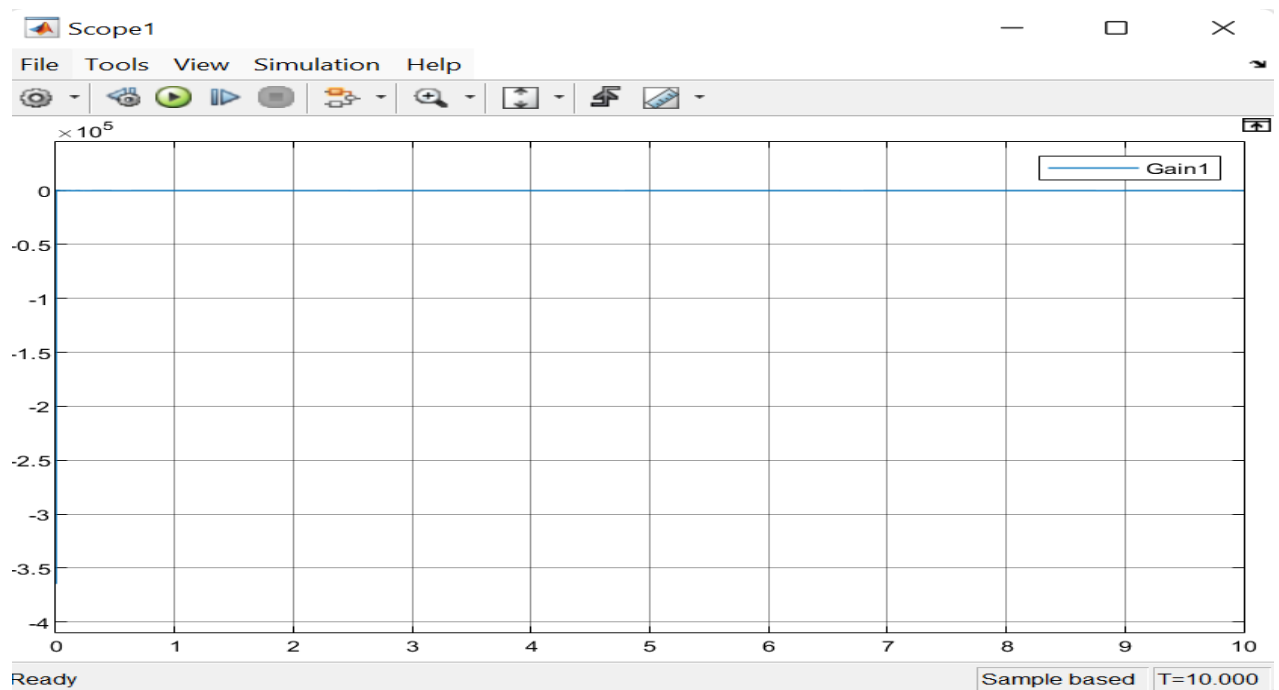


Figure 3. Control signal u

Now let's find the H_∞ norms for the given transfer functions,

$$C1(Is - (A + BK))^{-1}B_f$$

$$C2(Is - (A + BK))^{-1}B_f$$

$$(Is - (A + BK))^{-1}B_f$$

Where,

$$C1 = [1 \ 0], \quad C2 = [0 \ 1]$$

For that we will plot the singular values of the system to see the worst case gain and then check the Hinfnorm values to get the peak (supremum) of the largest singular value, we will calculate the peak gain of the system and the frequency at which the peak gain occurs.

```
A = [3 -1; -2 2];
B = [8;8];
Bf =[0;7];
Q = [9 0; 0 7];
gamma = 15.685;
P = are(A,B*B'-Bf*gamma^(-2)*Bf',Q);
K= -B'*P;
eig_=eig(A+B*K);
%disp(eig_)
%disp(K)

C1 = [1 0];
C2 = [0 1];

tf1 = tf(ss(A+B*K,Bf,C1,0));
tf2 = tf(ss(A+B*K,Bf,C1,0));
tf3 = tf(ss(A+B*K,Bf,eye(2),0));

sigma(tf1)
sigma(tf2)
sigma(tf3)
[ninf1,fpeak1]=hinfnorm(tf1);
[ninf2,fpeak2]=hinfnorm(tf2);
[ninf3,fpeak3]=hinfnorm(tf3);
disp([ninf1,fpeak1]);
disp([ninf2,fpeak2]);
disp([ninf3,fpeak3]);
```

Figure 4. Calculating the transfer function and H-infnorm values

For the transfer function ,

$$C1(Is - (A + BK))^{-1}B_f$$

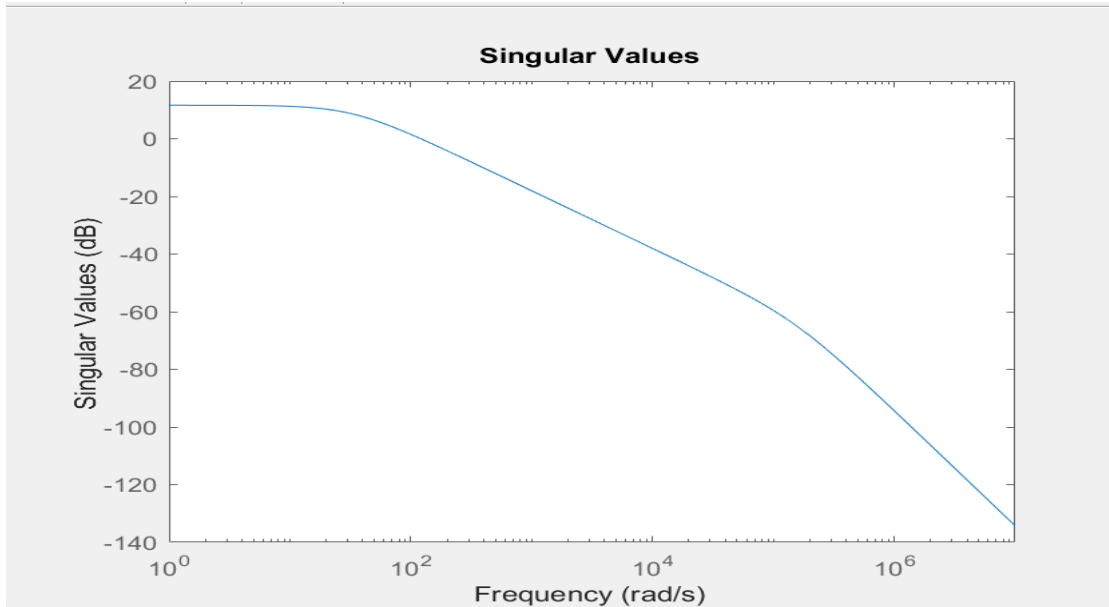


Figure 5. Singular values for tf1

The peak gain of the system and the frequency for peak gain are as,

$$[ninf, fpeak] = [3.8197, 0]$$

For transfer function,

$$C2(Is - (A + BK))^{-1}B_f$$

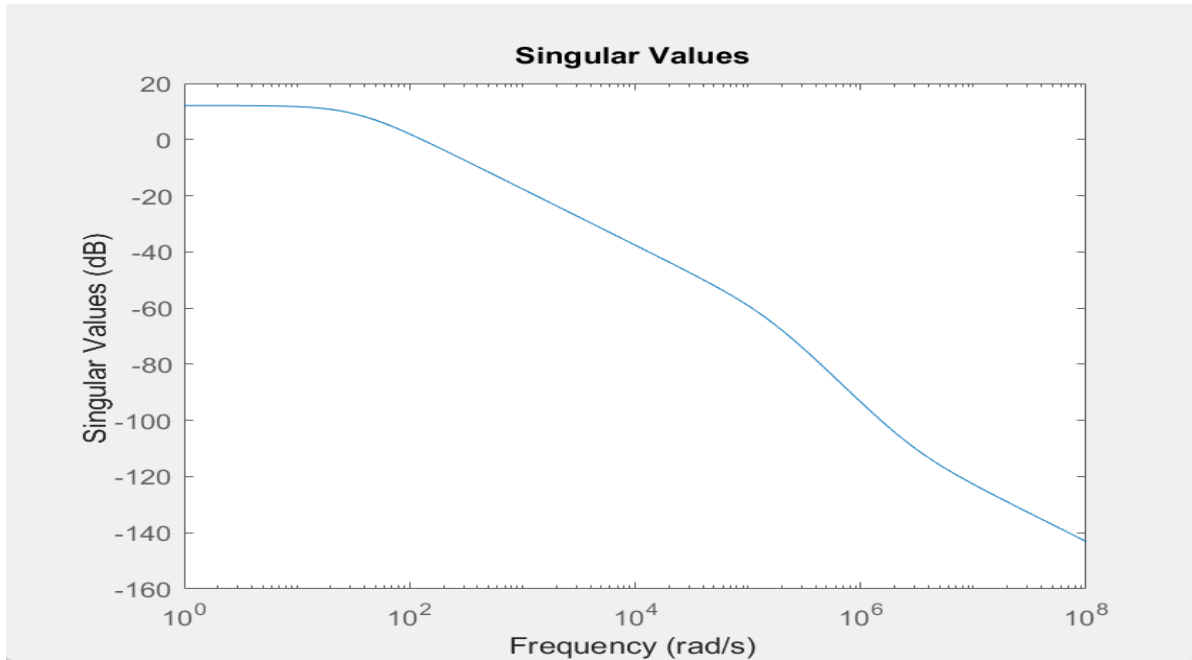


Figure 6. Singular values for tf2

The peak gain of the system and the frequency for peak gain are as,

$$[ninf, fpeak] = [4.0328, 0]$$

For transfer function,

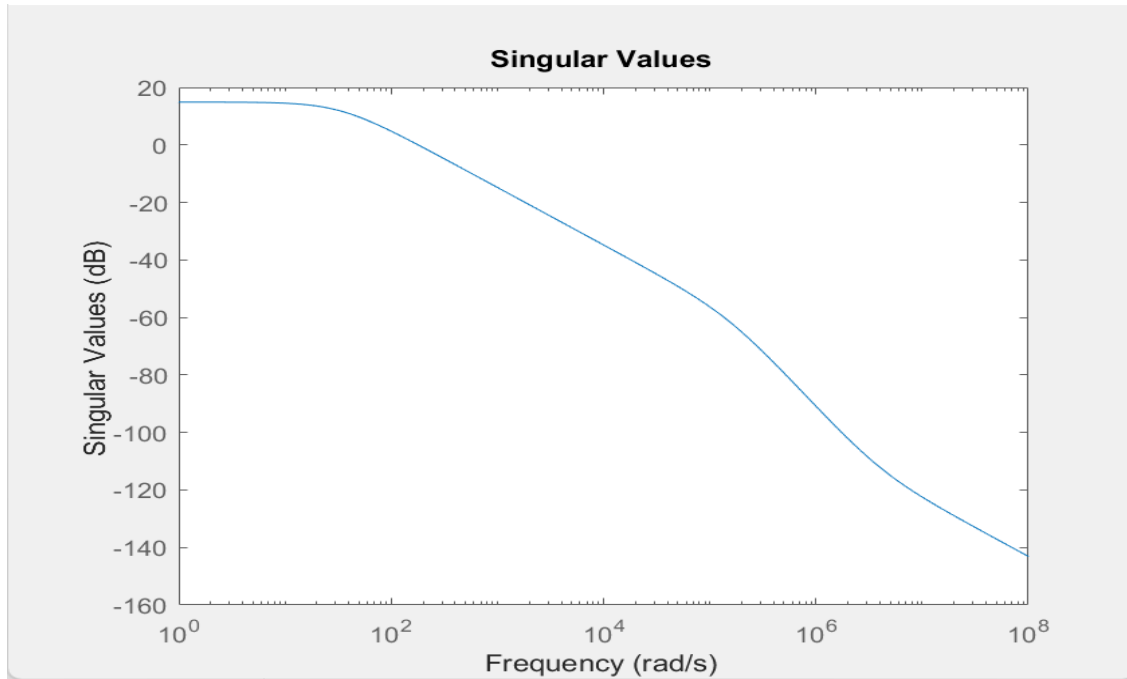


Figure 7. Singular values for tf3

The peak gain of the system and the frequency for peak gain are as,

$$[ninf, fpeak] = [5.5546, 0]$$

3. Conclusion

In this laboratory work we have modeled a disturbed linear plant in the presence of multi harmonic disturbance signal, and then synthesized an optimal controller for that plant. Moreover we have also calculated the H-inf norms for certain transfer functions of our state space model to calculate the worst gain scenario for the disturbed plant.